Rajdeep Dhar

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Professional Summary

Information Technology undergraduate with strong DSA fundamentals and expertise in systems and backend development. Proficient in engineering highly concurrent applications using C++ and developing resilient, scalable web architectures with the MERN stack. Experienced in containerizing and deploying systems with Docker on cloud platforms, focusing on optimizing system performance, reducing latency, and delivering maintainable code across complex software pipelines.

Education

Jadavpur University *Kolkata, India | Bachelor of Engineering in IT (CGPA: 9.33/10)*

Class Rank: 1 | CP Team Member | Coursework: DBMS, OOP, DSA, Computer Networks

Projects & Experiences

ResumeRise *Self Project | Mar 2026 - Jun 2026*

- Developed a 4-stage async Express middleware pipeline utilizing a Context Object pattern, enabling per-step error recovery and stable request lifecycles for concurrent users.
- Reduced AI token consumption by approximately 5,000 tokens and decreased response latency by around 5 seconds per request.
- Introduced a caching layer that deduplicates job scrapes and resume parses using URL matching and Mongoose referencing.
- Engineered a 4-tier Gemini model fallback chain with Zod schema validation across all tiers, ensuring uninterrupted report generation.
- Enhanced resume-JD alignment accuracy by approximately 30% by classifying AI-extracted job skills into priority tiers using Zod contracts.

Technologies: React, Node.js, Express, MongoDB, Zod, Puppeteer, Cheerio, Tailwind CSS

Multithreaded Reverse Proxy Server *Self Project | Jun 2026 - Jun 2026*

- Engineered a 16-shard LRU cache that achieved 3.53M ops/sec under 24-thread concurrent load, utilizing per-shard O(1) lookups and TTL expiration.
- Mitigated thundering herd cache stampedes in concurrent request handling by implementing a request coalescer for HTTP/1.1 GET requests.
- Developed a 24-thread pool for concurrent connection dispatch, removing per-connection spawn overhead and enabling deadlock-free shutdown.
- Containerized and deployed the proxy server to Render via a multi-stage Dockerfile that compiles the CMake project inside a GCC build image.

Technologies: C++, CMake, sockpp, cpp-httplib, OpenSSL, picohttpparser, multithreading, networking

Skills

Databases: MySQL, SQLite, MongoDB

Frameworks & Libraries: React, Express, React Router, OpenSSL, Zod

Programming Languages: C, C++, Java, JavaScript

Tools & Platforms: Git, GitHub, Vite, CMake, Linux, Docker